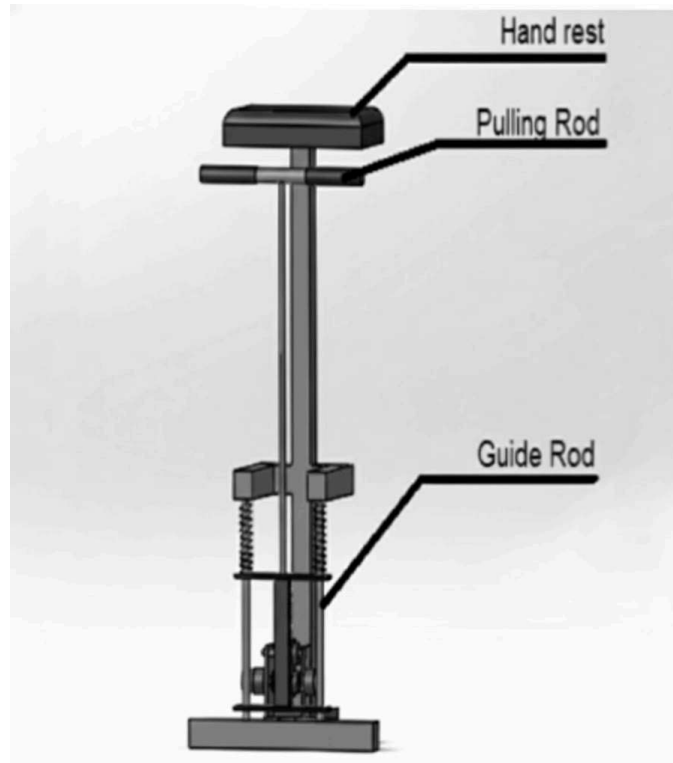


TA212: Manufacturing processes - II
1st Sem 2025-26
Group No: 19

AnthroVolt



Akshat Kasarwal	Priyanshu Joshi	Aayushman Tripathi	Harshit Dashore
240085	240806	240025	240439

Sarthak Bharti
240938

Bhargavi Borse
240284

Shambhavi Singh
240958

Guide: Mr. R.N. Tudu
Lab I/c.: Mr. Aman Singh

Tutor: Mr.
Course Instructor: Dr. Sarvesh Mishra

Index

S. No.	Content	Page No.
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3.	Acknowledgment	5
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5.	Motivation	7
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7.	Log Sheets	20
8.	Cost Estimation	28

Parts List

S. No.	Part Name	Quantity	Dimensions (mm)
1.	Hand Support	1	100x250x10
2.	Pulling Rod	1	250x300
3.	Guide Rod	2	400x50x20
4.	Spring	2	25 diameter (unstretched)
5.	Pinion	1	OD - 33 Teeth - 20
6.	Rack	1	115x50
7.	DC Motor	1	-
8.	Base Plate	1	350x350
9.	Support Rod	2	200x25
10.	Pulling Rod Support Rod	1	250x50
11.	Support Strip	3	250x20x5
12.	Angle 1	4	45x45x45
13.	Angle 2	2	45x45x20

Project Overview

This report outlines the design and operation of a vertical resistance device intended for upper-body training, rehabilitation, and potential **power generation**. The apparatus features a central column with key user components: a **"Hand rest"** for support and a **"Pulling Rod"** for performing the exercise.

Operationally, the user pulls the "Pulling Rod" upward, lifting a carriage mechanism. This carriage's motion is governed by vertical **"Guide Rods"** and resisted by a spring-loaded system, ensuring a smooth and controlled path for safety and efficacy. The base integrates **electromechanical components** to modulate resistance levels, **capture user-generated power**, and **provide real-time performance feedback**. This document will cover the detailed specifications and analysis of this electromechanical design.

Acknowledgment

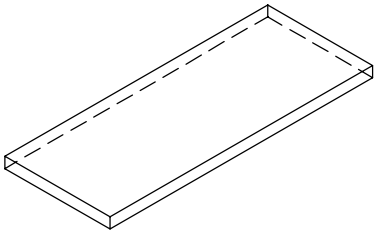
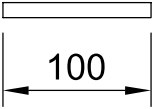
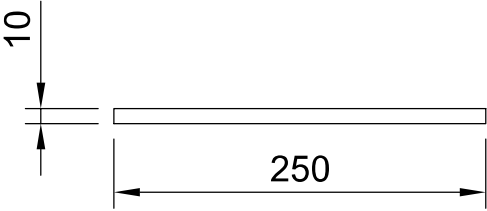
We would like to express our sincere gratitude to all those who have supported and guided us throughout the completion of this project. We are especially thankful to our guides and tutors for their valuable suggestions, encouragement, and feedback at various stages of the work. Their insights have greatly helped us in improving the quality of this project. We also acknowledge the use of various resources, references, and tools that provided us with the necessary knowledge base to carry out this work. Finally, we are grateful to each other as members of Group F19 for the teamwork, dedication, and cooperation that made the successful completion of this project possible.

Abstract

This report presents the design, operation, and analysis of a novel vertical resistance device for upper-body strength training, rehabilitation, and potential **power generation**. The design features a central guided carriage system activated by a **"Pulling Rod"** and stabilized by a **"Hand rest."** Motion is constrained by **"Guide Rods"** and opposed by a spring-loaded mechanism. The base integrates electromechanical components designed to modulate resistance, **capture user-generated power**, and **provide real-time performance feedback**. The primary objective is to document the complete electromechanical design, operational principles, and performance analysis, providing a technical overview of the device's specifications and its dual efficacy in exercise and energy harvesting.

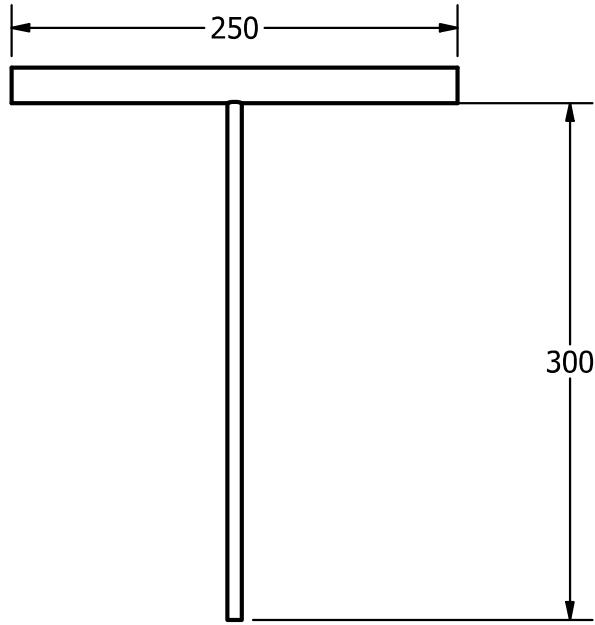
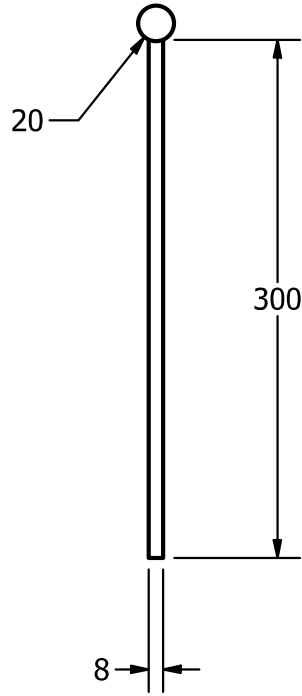
Motivation

The motivation for this project stems from a convergence of needs in modern health, sustainability, and technology. There is a persistent demand for safe, measurable devices for physical rehabilitation, alongside accessible equipment for general fitness. Simultaneously, there is a growing interest in sustainable energy solutions. Conventional exercise machines dissipate user-generated energy as heat; this project seeks to innovate by capturing and utilizing this human power. Furthermore, the integration of smart, data-driven feedback is becoming essential for user engagement and optimizing training. This project aims to address these areas by creating a single device that combines controlled exercise with energy harvesting and real-time performance monitoring.

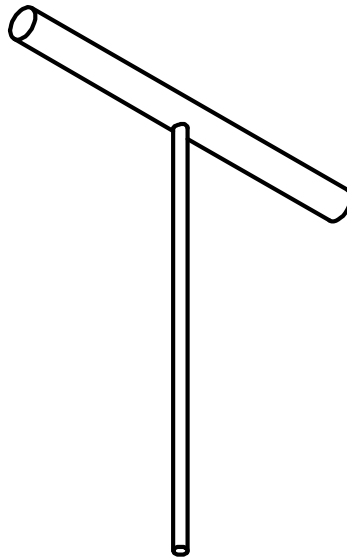


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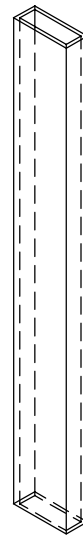
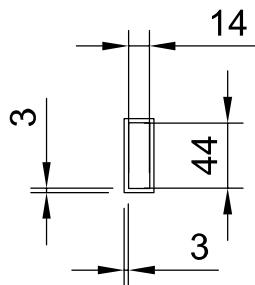
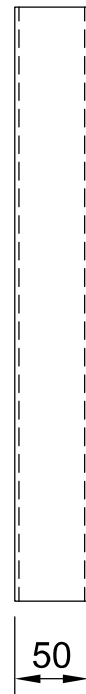
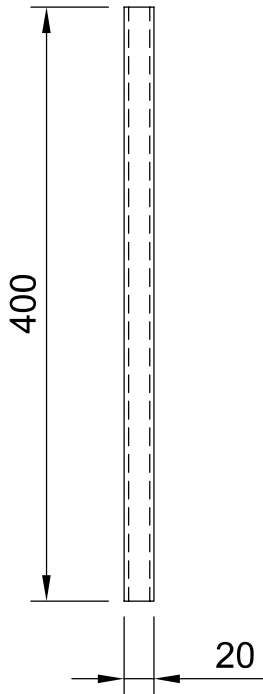
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		Title hand support	DWG No.		
			Rev.	Date of issue	Sheet 1/1



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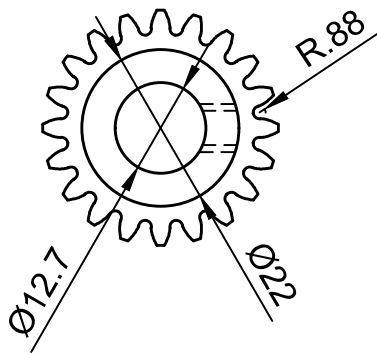
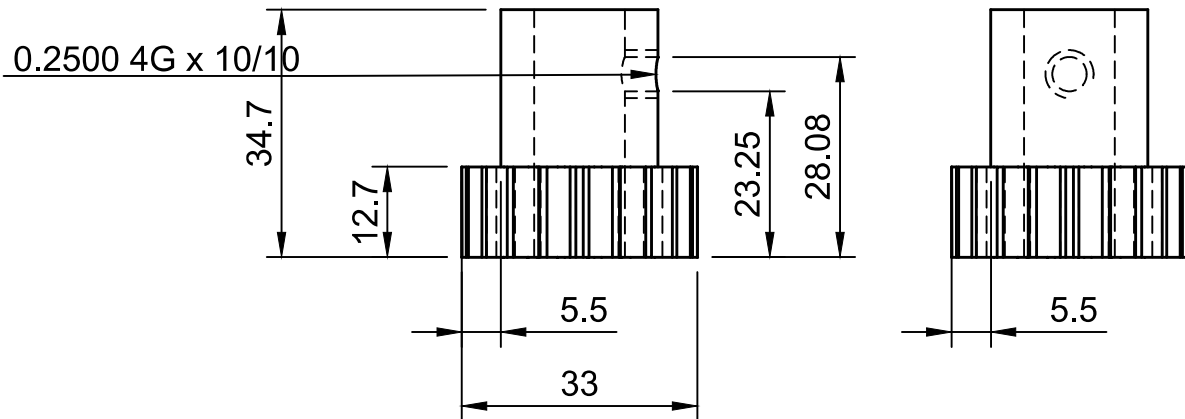


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QA								
MFG								
APPROVED								
			SIZE		DWG NO			REV
			A4					
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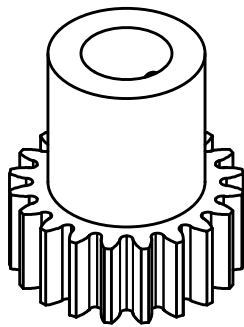
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		Title guide rod	DWG No.		
			Rev.	Date of issue	Sheet 1/1

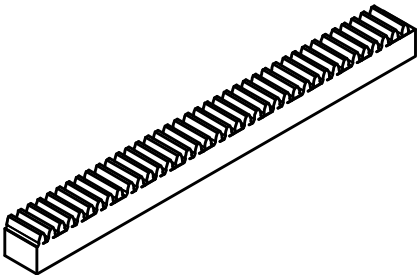
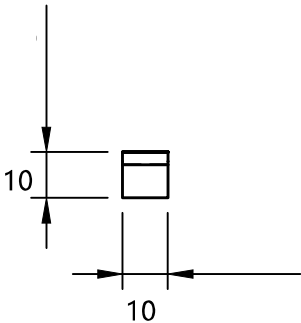
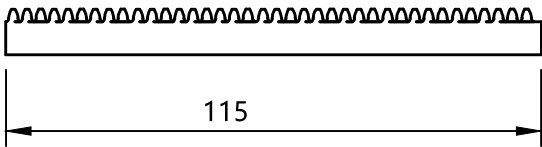


No of Teeth (n): 20
 Module (m): 1.5
 Pitch Diameter (m $\times$ n): 30.00 mm
 Outer Diameter OD = m(n+2): 33.00 mm
 Hole Diameter (ID): 12.7 mm
 Pressure Angle: 20°
 Root Fillet Radius: 0.880 mm
 Backlash: 0.00 mm
 Depth of Cut (2.157 $\times$ m): 3.2355
 Indexing (40 / n): 2
 Tap Hole Size: 5.2 mm drill & 1/4" tapping

SCALE: 1:1

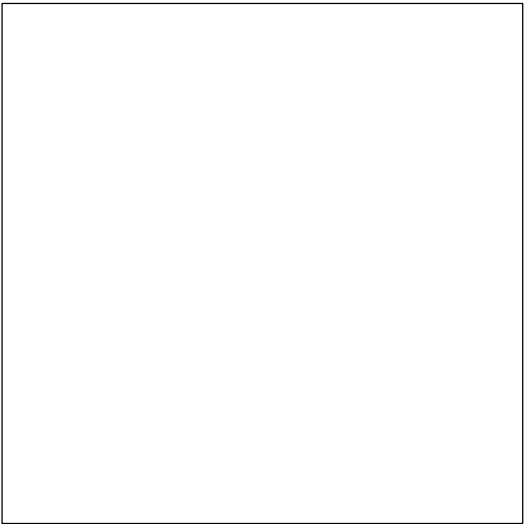
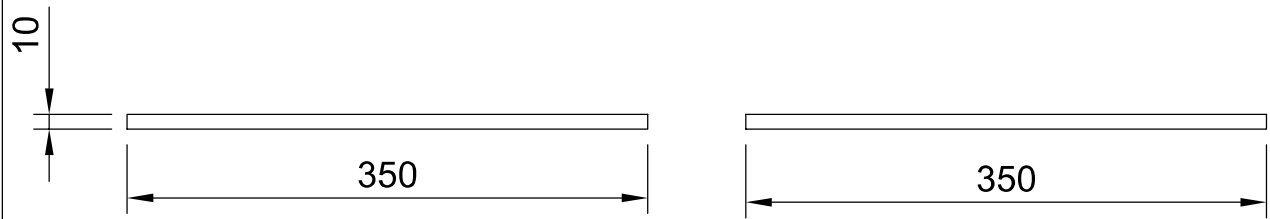


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		Title gear	DWG No.	
			Rev.	Date of issue
			Sheet 1/1	

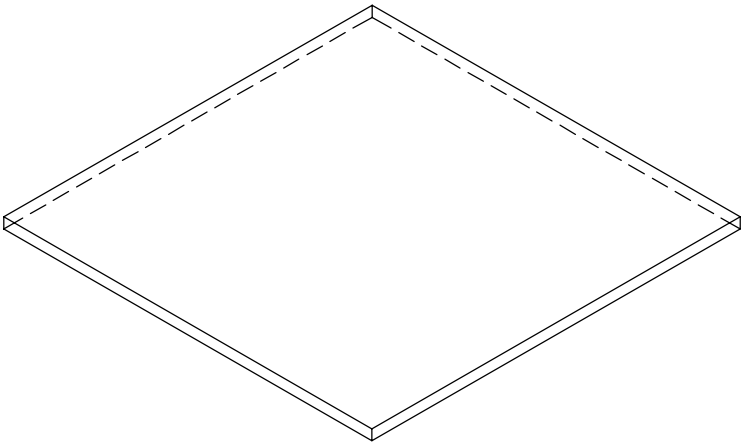


SCALE: 1:2
Module (m): 4.7
Pitch (p=nxm): 14.765
Length of rack (overall): 115 mm
Height of rack(overall): 50 mm
Tooth Height: 3.494mm
Pressure Angle: 20 degree

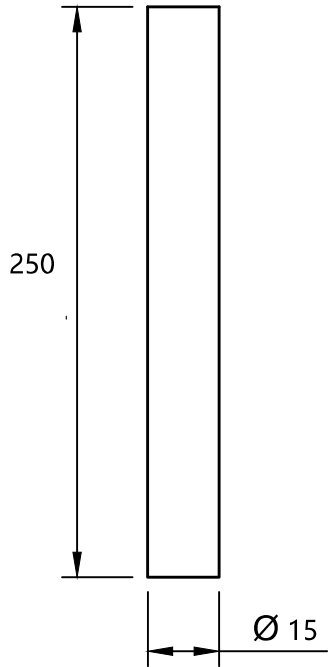
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		Title rack gear	DWG No.	
			Rev.	Date of issue
				Sheet 1/1



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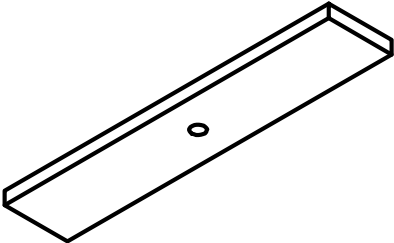
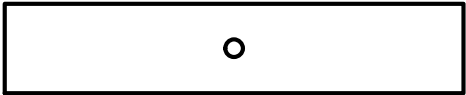
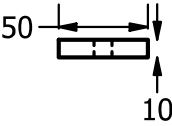
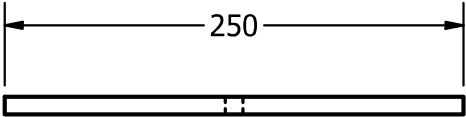
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		Document type	Document status		
		Title base plate	DWG No.		
			Rev.	Date of issue	Sheet 1/1



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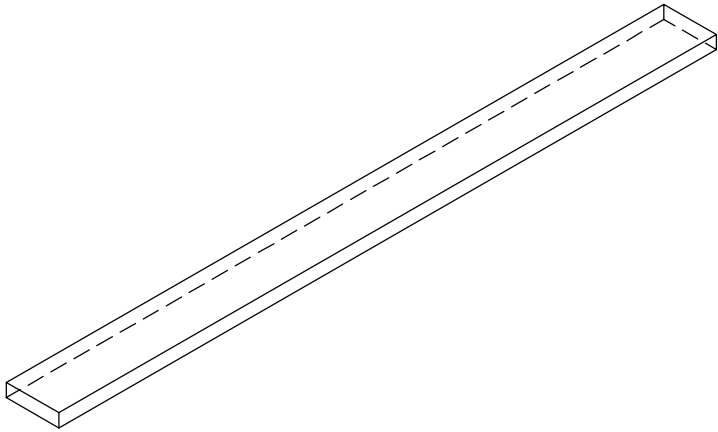
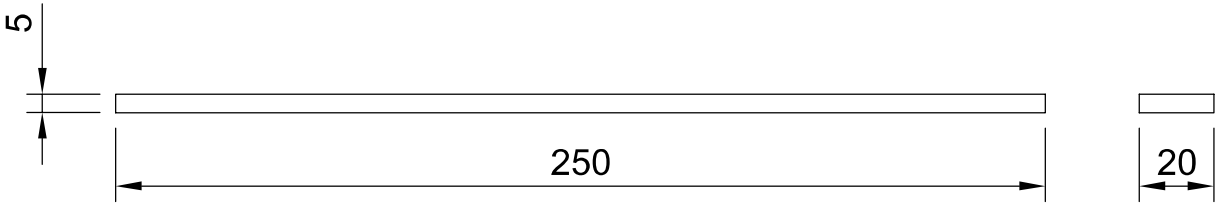


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		Document type	Document status	
		Title support rod	DWG No.	
			Rev.	Date of issue
			Sheet 1/1	



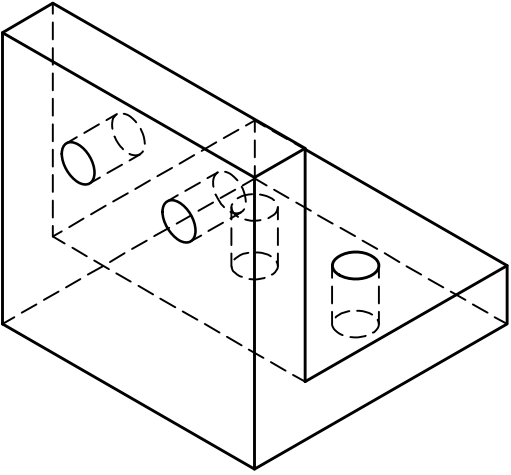
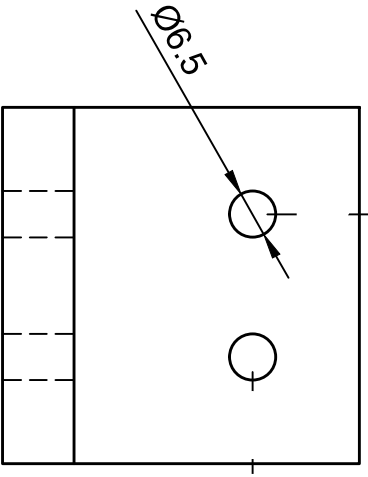
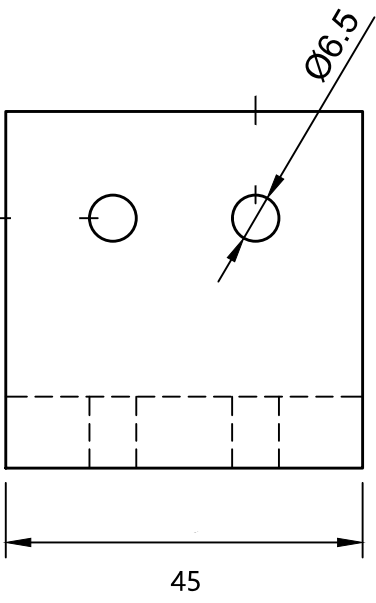
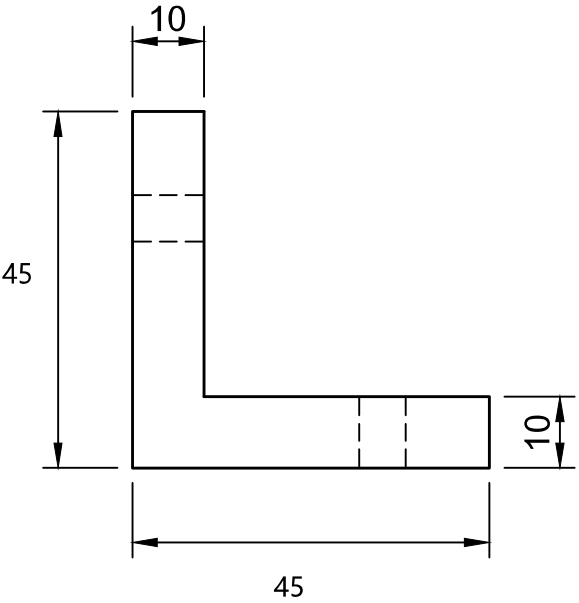
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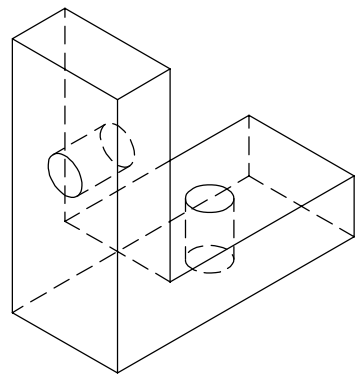
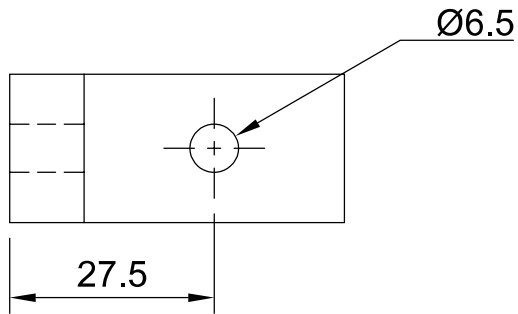
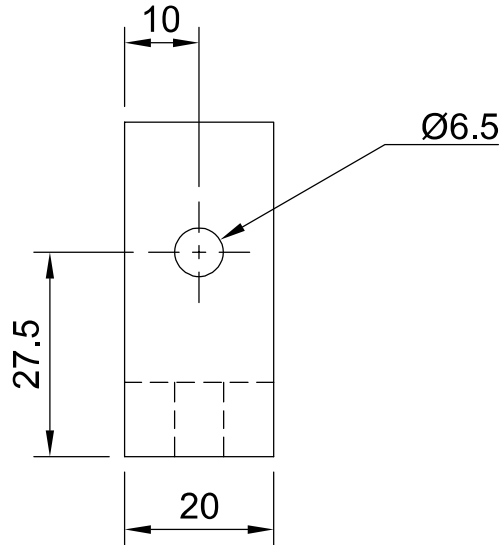
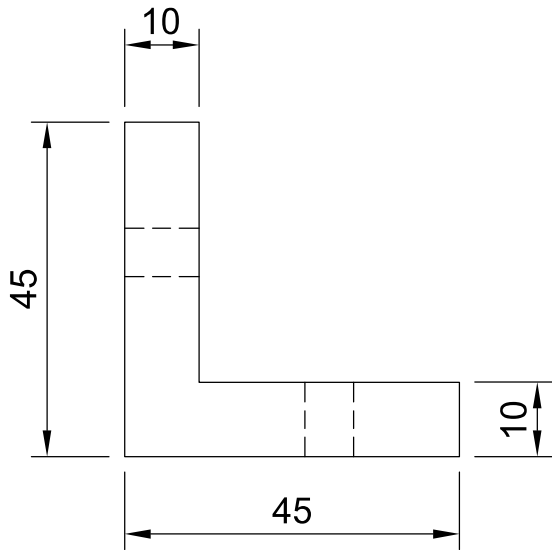
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			Rev.	Date of issue	Sheet 1/1



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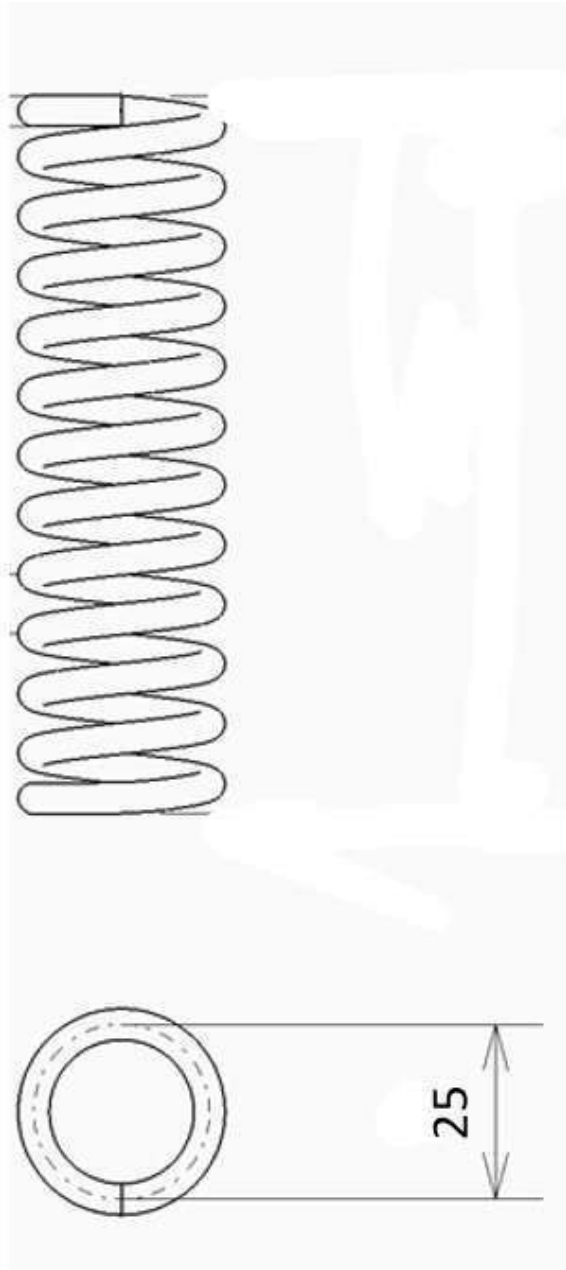
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		Document type	Document status	
		Title angle	DWG No.	
			Rev.	Date of issue
			Sheet 1/1	



SCALE: 1:1

Dept.	Technical reference	Created by Shambhavi Singh 11-11-2025	Approved by		
		Document type	Document status		
		Title angle 2	DWG No.		
			Rev.	Date of issue	Sheet 1/1

SPRING



Log Sheet - To be submitted by Individual Students

Title of the Project: Arduino Volt

Group No.: 19

Name: AAYUSHMAN TRIPATHI Roll No.: 240025

Section:

(1) No. of parts fully completed from the planned and approved main drawing (As per the index submitted earlier)

Project Lab No.	Date	Part Name	Work done	Guide Sign.
1 st	10/09/25			
2 nd	24/09/25	Filing of Square Slab pipe Filing of Circular Rod filing.		<i>Shubh</i> 24/9/25
3 rd	08/09/25	Cutting and Filing of Rod drilling of Slab		<i>Shubh</i> 08/09/25
4 th	15/10/25	rack	cutting rack for motor drilling filing	<i>Shubh</i> 15/10/25
5 th				
6 th				
7 th				
8 th				

(2) No. Parts assembled

Project Lab No.	Date	Name of Parts Assembled	Work done	Guide Sign.
1 st	29/10	rack supply	Assembly	<i>Shubh</i> 29/10/25
2 nd	1/11	Arduino	Code/assembly	<i>Shubh</i> 1/11/25
3 rd				
4 th				
5 th				
6 th				

Log Sheet - To be submitted by Individual Students

Title of the Project: *Andro Volt*

Group No.: 19

Name: *AKSHAT KASARWAL*

Roll No.: *240085*

Section:

(1) No. of parts fully completed from the planned and approved main drawing (As per the index submitted earlier)

Project Lab No.	Date	Part Name	Work done	Guide Sign.
1 st	10/09/25			
2 nd	24/09/25	Angle & support rod	Fixing Nut-bolt on Base plate	<i>[Signature]</i> 24/9/25
3 rd	06/10/25	Support rod & Angle, Joint b/w rod & angle	Fixing, drilling on rod sample	<i>[Signature]</i> 06/10/25
4 th	15/10/25	Support rod & Angle, Rag support	Assembly, drilling, filing	<i>[Signature]</i> 15/10/25
5 th	22/10/25	Assembly of support rod, Rag support and structure.	filing, drilling, cutting	<i>[Signature]</i> 22/10/25
6 th				
7 th				
8 th				

(2) No. Parts assembled

Project Lab No.	Date	Name of Parts Assembled	Work done	Guide Sign.
1 st	22/10/25	support rod, pin, rack and side plate	Assembly	<i>[Signature]</i> 22/10/25
2 nd				
3 rd				
4 th				
5 th				
6 th				

Log Sheet - To be submitted by Individual Students.

Title of the Project: ANTHROVOLT

Group No.: 19

Name: BHARGAVI BORSE

Roll No.: 240284

Section:

(1) No. of parts fully completed from the planned and approved main drawing (As per the index submitted earlier)

Project Lab No.	Date	Part Name	Work done	Guide Sign.
1 st	10/09/25			
2 nd	24/09/25	^{final} Drawing, Support & Pulling Rod	Cutting, Filing edges support	<i>[Signature]</i> 24/09/25
3 rd	8/10/25	Pinion gear *2	Lathe, surfacing teeth cutting	<i>[Signature]</i> 8/10/25
4 th				
5 th	22/10/25	Assembly of rack	threading, filing, cutting, assembly	<i>[Signature]</i> 22/10/25
6 th				
7 th				
8 th				

(2) No. Parts assembled

Project Lab No.	Date	Name of Parts Assembled	Work done	Guide Sign.
1 st	22/10/25	rack	assembly	<i>[Signature]</i> 22/10/25
2 nd	29/10	rack & support	assembly	<i>[Signature]</i> 29/10/25
3 rd	1/11	spring, rack, report	assembly, turning	<i>[Signature]</i> 01/11/25
4 th				
5 th				
6 th				

Log Sheet - To be submitted by Individual Students

Title of the Project: Anthro Volt

Group No.: 19

Name: KARSHIT DASHORE

Roll No.: 240439

Section:

(1) No. of parts fully completed from the planned and approved main drawing (As per the index submitted earlier)

Project Lab No.	Date	Part Name	Work done	Guide Sign.
1 st	10/9/25			
2 nd	24/9/25	fixing nut & bolt Base plate: assembly , square and circular rods cutting, filing	Cutting and Filing	<u>Pludin</u> 24/09/25
3 rd	8/10/25	Threading & fixing nut bolt in Spring support rod + filing of Parts	Various parts made	Pludin 15/10/25
4 th	15/10/25	supported handle rod support	assembly drilling cutting	<u>Pludin</u> 15/10/25
5 th	22/10/25	Assembly rack	cutting, threading	<u>Pludin</u> 22/10/25
6 th				
7 th				
8 th				

(2) No. Parts assembled

Project Lab No.	Date	Name of Parts Assembled	Work done	Guide Sign.
1 st	22/10/25	1 rack with base plate	assembly	<u>Pludin</u> 22/10/25
2 nd	29/10/25	Rack support	Assembly	<u>Pludin</u> 29/10/25
3 rd	1/11/25	Complete assembly, Spring, rack fitting	Assembly	<u>Vinay</u> 01/11/25
4 th				
5 th				
6 th				

Log Sheet - To be submitted by Individual Students

Title of the Project: Antenna Volt

Group No.: 19

Name: PRIYANSHU JOSHI

Roll No.: 240806

Section:

(1) No. of parts fully completed from the planned and approved main drawing (As per the index submitted earlier)

Project Lab No.	Date	Part Name	Work done	Guide Sign.
1 st	10/09/25			
2 nd	24/09/25	Base Plate Assembly , Support Rods (fixing Nut & Bolt) ✓	Cutting & Filing	<u>Prady</u> 24/9/25
3 rd	08/09/25	Cutting and filing of Rod, Drilling fixing nut & bolts.	Cutting, Filing, Drilling	<u>Prady</u> 8/10/25
4 th	15/09/25	Drilling in Shaft, Gear, Assembly, Angle	Cutting, Filing, Drilling, Assembly	<u>Prady</u> 15/10/25
5 th				
6 th				
7 th				
8 th				

(2) No. Parts assembled

Project Lab No.	Date	Name of Parts Assembled	Work done	Guide Sign.
1 st				
2 nd	29/10/25	Rack Support	Assembly	<u>Prady</u> 29/10/25
3 rd	1/11/25	Rack fitting, Spring fitting	Assembly	<u>Prady</u> 1/11/25
4 th				
5 th				
6 th				

Log Sheet - To be submitted by Individual Students

Title of the Project:

Group No.: 19

Name: SARTHAK BHARTI Roll No.: 240938 Section:

(1) No. of parts fully completed from the planned and approved main drawing (As per the index submitted earlier)

Project Lab No.	Date	Part Name	Work done	Guide Sign.
1 st	8/10/25	Guide Rod Cutting & Filling(x2) Support Rod Filling	Assembly	<i>[Signature]</i> 22/10/25
2 nd	22/10/25	Rod Cutting, filling Support Drilling	Assembly	<i>[Signature]</i> 22/10/25
3 rd	1/11/25	Rod Cutting, Rack Cutting, Screws & nut fix	work done	<i>[Signature]</i> 02/11/25
4 th				
5 th				
6 th				
7 th				
8 th				

(2) No. Parts assembled

Project Lab No.	Date	Name of Parts Assembled	Work done	Guide Sign.
1 st	22/10/25	Support plate assembled	✓	<i>[Signature]</i> 22/10/25
2 nd	1/11/25	Assembly	✓	<i>[Signature]</i> 02/11/25
3 rd				
4 th				
5 th				
6 th				

Log Sheet - To be submitted by Individual Students

Title of the Project: ANTHROVOLT

Group No.: 19

Name: SHAMBHAVI SINGH

Roll No.: 240958

Section: ~~19~~

(1) No. of parts fully completed from the planned and approved main drawing (As per the index submitted earlier)

Project Lab No.	Date	Part Name	Work done	Guide Sign.
1 st	10/09/25			
2 nd	24/09/25	Drawing Rack & Pinion, Guide & Support Rod, Motor Support, angle, shaft	Drawing	Phedu 24/9/25
3 rd	08/10/25	Pinion gear x 2	Milling, teeth cutting, lathe	Phedu 8/10/25
4 th	15/10/25	Drilling , Shaft, gear, angle	Drilling, filing filing, Cutting, Assembly	Phedu 15/10/25
5 th				
6 th				
7 th				
8 th				

(2) No. Parts assembled

Project Lab No.	Date	Name of Parts Assembled	Work done	Guide Sign.
1 st				
2 nd	29/10/25	Rack & Support	Assembly	Phedu 29/10/25
3 rd	1/11/25	Spring assembly Report & Autocad, presentation	Assembly, Drawing Report	Phedu 1/11/25
4 th				
5 th				
6 th				

Log Sheet - To be submitted by Individual Students

Title of the Project: ARTURO VOLT

Group No.: 19

Name: VIKASH SINGH

Roll No.: 241119

Section:

(1) No. of parts fully completed from the planned and approved main drawing (As per the index submitted earlier)

Project Lab No.	Date	Part Name	Work done	Guide Sign.
1 st	10/09/25			
2 nd	24/09/25	Angle & support rod	joining Fitting base plate	<u>Shuchi</u> 24/09/25
3 rd	5/10/25	support rod & angle 1 joint 5th rod + angle	fixing plate drilling error	<u>Shuchi</u> 5/10/25
4 th	19/10/25	teeth rack	cutting rack distortion drilling	<u>Shuchi</u> 19/10/25
5 th	22/10/25	assembly	cutting drilling	<u>Shuchi</u> 22/10/25
6 th				
7 th				
8 th				

(2) No. Parts assembled

Project Lab No.	Date	Name of Parts Assembled	Work done	Guide Sign.
1 st	22/10/25	rack + base plate	Assembly	<u>Shuchi</u> 22/10/25
2 nd	29/10/25	rack support	Assembly	<u>Shuchi</u> 29/10/25
3 rd	1/11/25	Complete assembly / rack fitting	Assembly	<u>Shuchi</u> 1/11/25
4 th				
5 th				
6 th				

Cost Estimation

Category	Description	Quantity / Time	Rate (Rs.)	Amount (Rs.)
Material Cost	Mild Steel	17.43 Kg	90 / Kg	1568.70
	Nuts & Bolts	0.5 Kg	130 / Kg	65.00
	Clips, Springs & Misc.	1 Kg	120 / Kg	120.00
Subtotal (Material)				1753.70
Machine Usage Cost	Cutting	1.0 Hr	60 / Hr	60.00
	Drilling	2.0 Hr	100 / Hr	200.00
	Lathe	2.5 Hr	350 / Hr	875.00
	Milling	3.5 Hr	450 / Hr	1575.00
Subtotal (Machine Usage)		(Total 9 Hrs)		2710.00
Labor Cost	Skilled Labor	18 Hrs (2.25 Days*)	850 / Day	1912.50
Subtotal (Labor)				1912.50
Electronics Kit	Single Motor Kit	1 Set	2500 / Set	2500.00
Subtotal (Electronics)				2500.00
Total Estimated Cost				Rs. 8,876.20